

A
Major Project Stage – II
Report On
Enhancing Phishing Detection: A Novel
Hybrid Deep Learning Framework For
Cybercrime Forensics

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for the award of the degree of

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IN
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Certificate



Certified that this **Major Project Stage-II** Report entitled, **“Enhancing Phishing Detection: A Novel Hybrid Deep Learning Framework For Cybercrime Forensics ”** is the bonafide work of **Muthyam Pranavi (H.T.No. 22TR1A6639), Nagunuri Deekshitha (H.T.No. 22TR1A6640), Pathuri Harikesha (H.T.No.22TR1A6645) and Marri Yashwanth (H.T.No.22TR1A6634)** of IV Year, **CSE (AI & ML)** in the year 2026 in partial fulfillment of the requirements to award the Degree of Bachelor of Technology in **COMPUTER SCIENCE & ENGINEERING (AI & ML) Branch** of **Sree Chaitanya Institute of Technological Sciences**, Karimnagar.

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DECLARATION

We hereby declare that the work which is being presented in this report entitled “**Enhancing Phishing Detection: A Novel Hybrid Deep Learning Framework For Cybercrime Forensics**”, submitted towards the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in COMPUTER SCIENCE & ENGINEERING (AI & ML)**, Sree Chaitanya Institute of Technological Sciences, Karimnagar is an authentic record of our own work carried out under the supervision of **Mrs. K.SIREESHA, Assistant Professor, Department of CSE (AI & ML)**, Sree Chaitanya Institute of Technological Sciences, Karimnagar.

To the best of our knowledge and belief, this project bears no resemblance with any report submitted to Sree Chaitanya Institute of Technological Sciences or any other University for the award of any degree or diploma.

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ABSTRACT

Protecting against interference is essential at a time when wireless communications are essential for sending large amounts of data. Our research presents a novel deep learning technique, the ResNeXt method and embedded Gated Recurrent Unit (GRU) model (RNT), rigorously developed for real-time phishing attack detection. Focused on countering the escalating threat of phishing assaults and bolstering digital forensics, our systematic approach involves SMOTE for managing data imbalance during initial data processing. The model's discriminative capability is improved, particularly in the feature extraction process, when autoencoders and ResNet (EARN) are integrated with feature engineering. The ensemble technique of feature extraction reveals crucial data patterns. At the core of our AI categorization is the RNT model, optimized using hyperparameters through the Jaya optimization method (RNT-J). Rigorously tested on real phishing attack datasets, our AI model consistently outperforms state-of-the-art algorithms by a substantial margin of 11% to 19% while maintaining exceptional computing efficiency. Furthermore, our model achieves 98% accuracy, low false positive/false negative values, and a statistical execution time with a mean of 36.99s, median of 35.99s, minimum of 34.99s, maximum of 41.99s, and a standard deviation of 1.10s. Moreover, it demonstrates superior accuracy with SMOTE (98%) and without SMOTE (83%) compared to other algorithms. This state-of-the-art AI study, which focuses on digital forensics, offers enhanced security and optimized productivity for businesses and industries, signifying a breakthrough in the continuing battle against phishing attempts. Through strengthening protection against interference in wireless communication, our AI research strives to amplify data accessibility, resilience, and trustworthiness in the face of cybersecurity threats within the organizational context.

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CHAPTER-1

INTRODUCTION

Phishing, a fraudulent activity utilizing both social and technological tactics to acquire financial and personal information from unsuspecting customers illicitly, remains a prevalent cybercrime. Among the well-established methods is email spoofing, involving the creation of deceptive emails with a forged origin, often distributed through social media platforms, impersonating reputable entities to trick users into visiting fraudulent websites and disclosing sensitive information like usernames and passwords. The utilization of devices by hackers to install malicious software further compounds the risk, facilitating unauthorized access and interception of user credentials. Phishers employ various platforms, including email, forums, URLs, messaging apps, text messages, and phone calls, to obtain user information. Their deceptive content often mirrors legitimate websites, enticing users to interact and divulge personal information. The primary aim of phishing is financial gain or identity theft, causing disruptions to businesses worldwide.

One important non-profit group that gathers, examines, and regularly updates the public on worldwide phishing activity is the Anti-Phishing Study Group. The surge in phishing attempts from 2022 to 2023, with over 300,000 recorded in July 2023, highlights the urgency of cybersecurity measures. Webmail remains a preferred target, with a significant increase from 600 to 1100 phishing attempts on well-known firms monthly, emphasizing the persistent threat. California's stringent laws, such as the Anti-Phishing Act of 2016, aim to penalize individuals involved in phishing assaults with imprisonment for up to five years or fines.

Criminals resort to creating illicit replicas of legitimate websites and communications, especially from financial institutions, deploying misleading emails to a broad audience. This phishing tactic, involving logos and phrases from reputable businesses, catches users off guard, leading them to counterfeit websites and increasing the risk of data misuse. Despite the urgency of phishing prevention, many organizations need more technology for detecting fraudulent URLs. Deep Learning (DL) algorithms, including Graph Convolutional Networks (GCN) and Bayesian Addition Regression Trees (BART), have shown promise in identifying features in observed datasets.

Traditional URL detection relies on blocklists, but their effectiveness is challenged by the constant rise of malicious URLs not on the list, facilitated by methods like the Domain Generation Algorithm (DGA). Recent studies in Phishing Detection (PD) focus on DL methods, with models like XGBoost for text preprocessing in email bodies and URLs. XGBoost, by examining email structures, web addresses, files, sender information, and metadata, efficiently processes large databases, extracting essential features and managing noise for effective phishing categorization.

Modern procedures necessitate a departure from traditional methods, incorporating a higher degree of human intervention. Modelling approaches using robust chronological datasets, including Graph Convolutional Networks (GCN), Recurrent Neural Networks (RNN), and Artificial Neural Networks (ANN), significantly enhance the efficacy of phishing detection techniques. In the realm of digital forensics, these neural network models play a crucial role in identifying and preventing phishing attempts, contributing to a more secure cyber field.

CHAPTER-2

PROJECT ANALYSIS

2.1 SYSTEM REQUIREMENTS

HARDWARE REQUIREMENTS :-

- **Processor** - Pentium –IV
- **RAM** - 4 GB (min)
- **Hard Disk** - 20 GB
- **Key Board** - Standard Windows Keyboard
- **Mouse** - Two or Three Button Mouse
- **Monitor** - SVGA

SOFTWARE REQUIREMENTS:-

- ❖ **Operating system** : Windows 7 Ultimate.
- ❖ **Coding Language** : Python.
- ❖ **Front-End** : Python.
- ❖ **Back-End** : Django-ORM
- ❖ **Designing** : Html, css, javascript.
- ❖ **Data Base** : MySQL (WAMP Server).

EXISTING SYSTEM

Author in outlined a method for identifying phishing attacks based on URL analysis. Their research used many algorithms to examine URLs from a range of data sets, comparing results using different deep learning (DL) techniques and hierarchical structures. To determine if this approach was effective, certain URL characteristics had to be evaluated, the website's administration and functionality had to be verified, and the website's visual representation had to be evaluated.

DL algorithms were used with skill to examine various aspects of URLs and web pages. Paradoxically, the author in presented a unique technique that focuses on accurate and successful phishing website identification using URL analysis. We define our novel neural network (NN) architecture in terms of many parallel components, the first of which is the removal of surface-level URL characteristics. However, in the work of, scholars were able to use simple characteristics to determine the legality of URLs while still producing accurate and trustworthy deep features of URLs. The findings of each component are combined to establish the maximum efficiency of the technology. A thorough analysis of a dataset from the internet confirms that, even after devoting a reasonable amount of time to the identification of phishing websites, our system is still comparable to other detection algorithms.

Researchers have developed artificial intelligence (AI)-based algorithms that employ machine learning and deep learning approaches in response to the widespread cyber dangers and vulnerabilities that internet users confront. The goal was to build a strong system that could identify phishing attempts and lessen the dangers of cyberattacks. The system aims at extracting this information from URLs by using a Convolutional Neural Network (CNN) using n-gram features.

2.2.1 DISADVANTAGES OF EXISTING SYSTEM

The researchers used sequencing techniques for reliable resource identification. Their suggested method showed a 95.38% TP rate in successfully identifying zero-day and phishing assaults.

Past research used the text structures of websites to build frameworks for phishing detection, but phishers managed to avoid detection by including information from outside sources.

Author in examined the effectiveness of the long short-term memory (LSTM) classifier while investigating the field of spoofing site prediction using hyperlinks as a data source for DL models.

Proposed System

Modern procedures necessitate a departure from traditional methods, incorporating a higher degree of human intervention. Modeling approaches using robust chronological datasets, including Graph Convolutional Networks (GCN), Recurrent Neural Networks (RNN), and Artificial Neural Networks (ANN), significantly enhance the efficacy of phishing detection techniques. In the realm of digital forensics, these neural network models play a crucial role in identifying and preventing phishing attempts, contributing to a more secure cyber field. This study introduces noteworthy contributions and novelty to the field of cyber security and phishing attack detection, emphasizing innovation, adaptability, and efficiency:

1) **Lightweight Model for Reduced Processing Time:** The research introduces a ResNeXt-embedded Gated

Recurrent Unit (RNT) model designed for real-time phishing attack detection. This lightweight strategy aim involves significantly reducing the time for processing, which raises defence system effectiveness.

2) By introducing the Jaya optimization approach (RNTJ) for tuning hyperparameters, the research raises the possibility of responses that are flexible through dynamic tuning. Optimizing the model dynamically

yields optimal values that enhance responsiveness and ensure its adaptability to a range of adaptive phishing attack scenarios.

3) The project presents a novel design that uses the SMOTE for information preparation for recognizing phishing attacks. The framework's resistance to various kinds of phishing behaviour scenarios is ensured through this new approach to data imbalanced.

4) Another significant property of the framework is its ability to adapt to new inputs readily. Such adjustable functionality gradually boosts the algorithm's reliability by ensuring that the recognition approach remains effective in confronting the effects of changing phishing techniques and attributes.

2.3.1 ADVANTAGES OF PROPOSED SYSTEM

1) Thorough Runtime Network Analysis: By utilizing EARN Ensemble (autoencoders and ResNet) to introduce sophisticated feature extraction, the study surpasses conventional metrics. The model's discriminative capability is improved by the ensemble technique, making it possible to identify important patterns linked to phishing assaults.

2) Improved Model Settings with RNT-J: The study advances the use of the Jaya optimization method (RNT-J) for model hyper parameter optimization. Through constant fine-tuning, the model is made to function at its best, which improves the accuracy of phishing detection.

3) Comprehensive Evaluation and Outperformance: Through rigorous testing on real phishing attack datasets, the proposed algorithm continuously outperforms the most advanced algorithms already in use. This significant contribution highlights the model's capacity to preserve computational efficiency while achieving better outcomes, ranging from an 11% to 19% improvement across several assessment measures.

2.4 PRELIMINARY INVESTIGATION

The first and foremost strategy for development of a project starts from the thought of designing a mail enabled platform for a small firm in which it is easy and convenient of sending and receiving messages, there is a search engine ,address book and also including some entertaining games. When it is approved by the organization and our project guide the first activity, ie. preliminary investigation begins. The activity has three parts:

- **Request Clarification**
- **Feasibility Study**
- **Request Approval**

REQUEST CLARIFICATION

After the approval of the request to the organization and project guide, with an investigation being considered, the project request must be examined to determine precisely what the system requires.

Here our project is basically meant for users within the company whose systems can be interconnected by the Local Area Network(LAN). In today's busy schedule man need everything should be provided in a readymade manner. So taking into consideration of the vastly use of the net in day to day life, the corresponding development of the portal came into existence.

2.5 SYSTEM STUDY

2.5.1 FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- ◆ ECONOMICAL FEASIBILITY
- ◆ TECHNICAL FEASIBILITY
- ◆ SOCIAL FEASIBILITY

ECONOMICAL FEASIBILITY

This project is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

TECHNICAL FEASIBILITY

This project is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

SOCIAL FEASIBILITY

The aspect of project is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

2.6 FEASIBILITY ANALYSIS

An important outcome of preliminary investigation is the determination that the system request is feasible. This is possible only if it is feasible within limited resource and time. The different feasibilities that have to be analyzed are

- **Operational Feasibility**
- **Economic Feasibility**
- **Technical Feasibility**

Operational Feasibility

Operational Feasibility deals with the study of prospects of the system to be developed. This system operationally eliminates all the tensions of the Admin and helps him in effectively tracking the project progress. This kind of automation will surely reduce the time and energy, which previously consumed in manual work. Based on the study, the system is proved to be operationally feasible.

Economic Feasibility

Economic Feasibility or Cost-benefit is an assessment of the economic justification for a computer based project. As hardware was installed from the beginning & for lots of purposes thus the cost on project of hardware is low. Since the system is a network based, any number of employees connected to the LAN within that organization can use this tool from at anytime. The Virtual Private Network is to be developed using the existing resources of the organization. So the project is economically feasible.

Technical Feasibility

According to Roger S. Pressman, Technical Feasibility is the assessment of the technical resources of the organization. The organization needs IBM compatible machines with a graphical web browser connected to the Internet and Intranet. The system is developed for platform Independent environment. Java Server Pages,

JavaScript, HTML, SQL server and WebLogic Server are used to develop the system. The technical feasibility has been carried out. The system is technically feasible for development and can be developed with the existing facility.

REQUEST APPROVAL

Not all request projects are desirable or feasible. Some organization receives so many project requests from client users that only few of them are pursued. However, those projects that are both feasible and desirable should be put into schedule. After a project request is approved, its cost, priority, completion time and personnel requirement is estimated and used to determine where to add it to any project list. Truly speaking, the approval of those above factors, development works can be launched.

CHAPTER 3

PROJECT DESIGN

SYSTEM DESIGN AND DEVELOPMENT

3.1 INPUT DESIGN

Input Design plays a vital role in the life cycle of software development, it requires very careful attention of developers. The input design is to feed data to the application as accurate as possible. So inputs are supposed to be designed effectively so that the errors occurring while feeding are minimized. According to Software Engineering Concepts, the input forms or screens are designed to provide to have a validation control over the input limit, range and other related validations.

This system has input screens in almost all the modules. Error messages are developed to alert the user whenever he commits some mistakes and guides him in the right way so that invalid entries are not made. Let us see deeply about this under module design.

Input design is the process of converting the user created input into a computer-based format. The goal of the input design is to make the data entry logical and free from errors. The error in the input are controlled by the input design. The application has been developed in user-friendly manner. The forms have been designed in such a way during the processing the cursor is placed in the position where must be entered. The user is also provided with an option to select an appropriate input from various alternatives related to the field in certain cases.

Validations are required for each data entered. Whenever a user enters an erroneous data, error message is displayed and the user can move on to the subsequent pages after completing all the entries in the current page.

3.2 OUTPUT DESIGN

The Output from the computer is required to mainly create an efficient method of communication within the company primarily among the project leader and his team members, in other words, the administrator and the clients. The output of VPN is the system which allows the project leader to manage his clients in terms of creating new clients and assigning new projects to them, maintaining a record of the project validity and providing folder level access to each client on the user side depending on the projects allotted to him. After completion of a project, a new project may be assigned to the client.

User authentication procedures are maintained at the initial stages itself. A new user may be created by the administrator himself or a user can himself register as a new user but the task of assigning projects and validating a new user rests with the administrator only.

The application starts running when it is executed for the first time. The server has to be started and then the internet explorer is used as the browser. The project will run on the local area network so the server machine will serve as the administrator while the other connected systems can act as the clients. The developed system is highly user friendly and can be easily understood by anyone using it even for the first time.

3.3 SYSTEM ARCHITECTURE

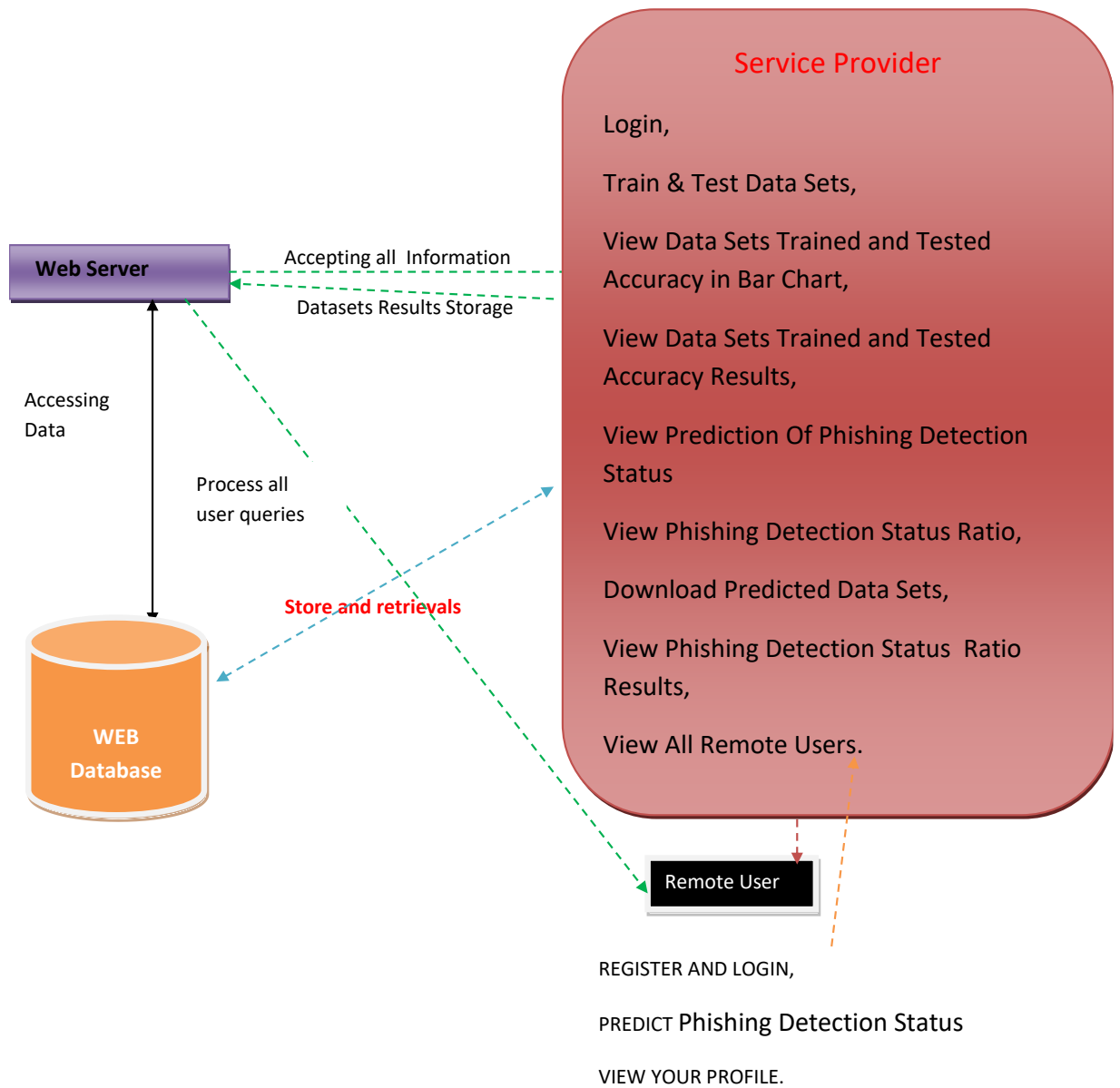


Fig:3.3 Architecture Diagram

3.4 Data Flow Diagram

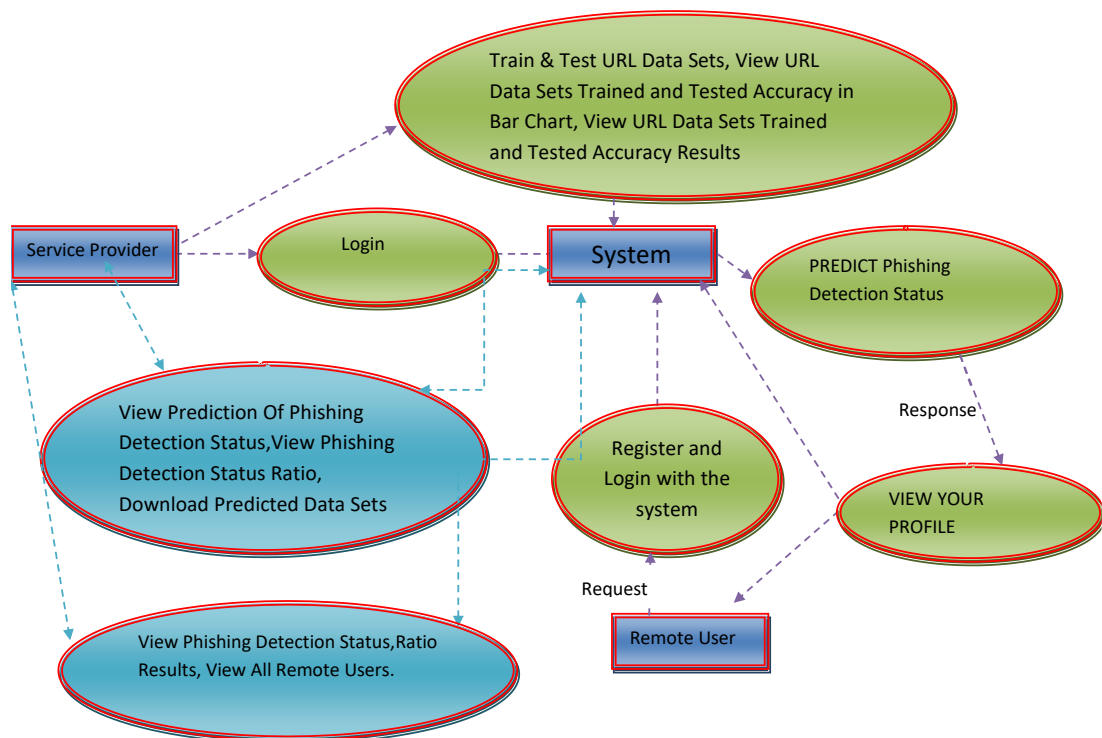


Fig:3.4 Data Flow Diagram

3.4.1 Flow Chart : Remote Use

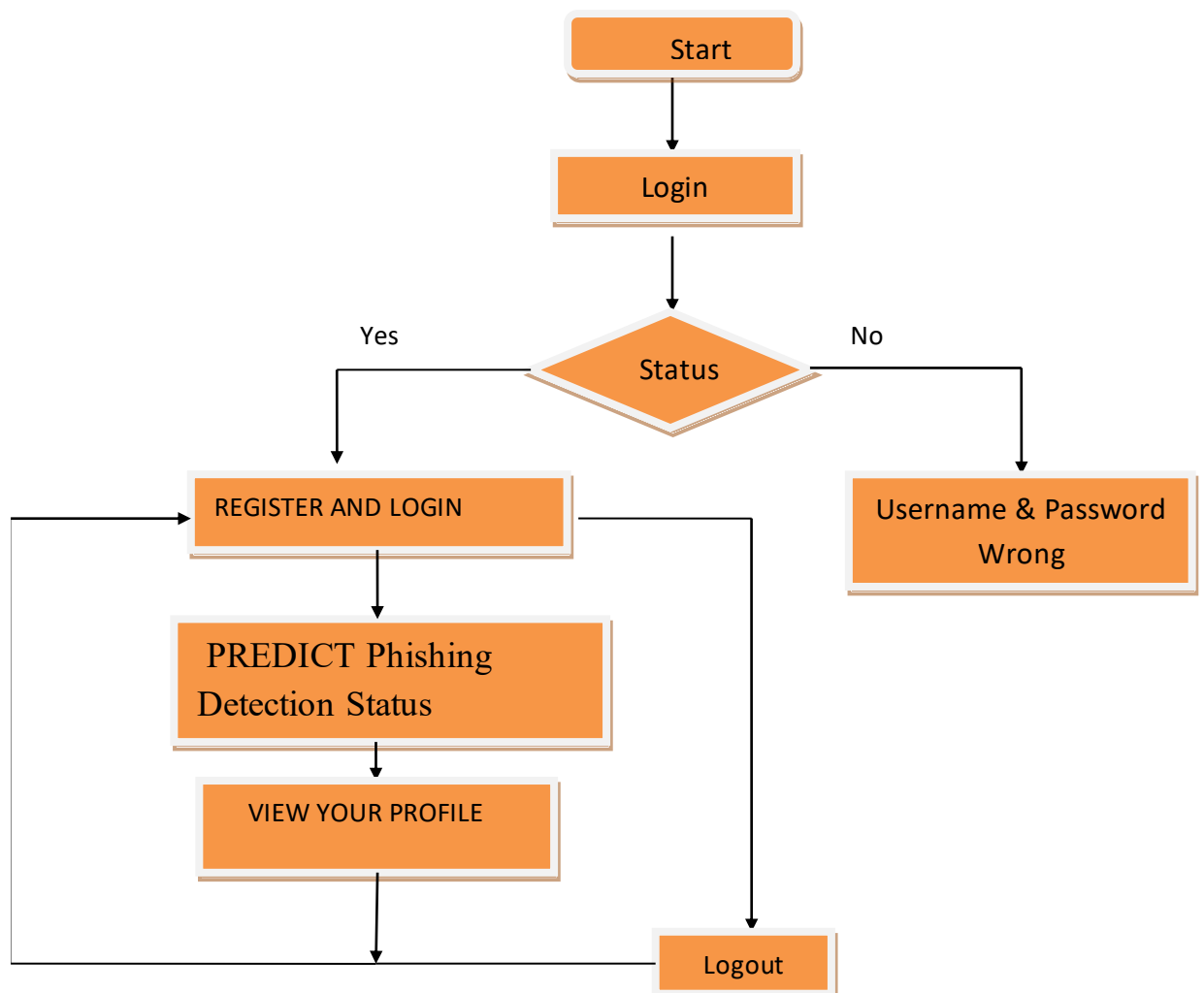


Fig:3.4.1 Flowchart For RemoteUser

3.4.2 Flow Chart : Service Provider

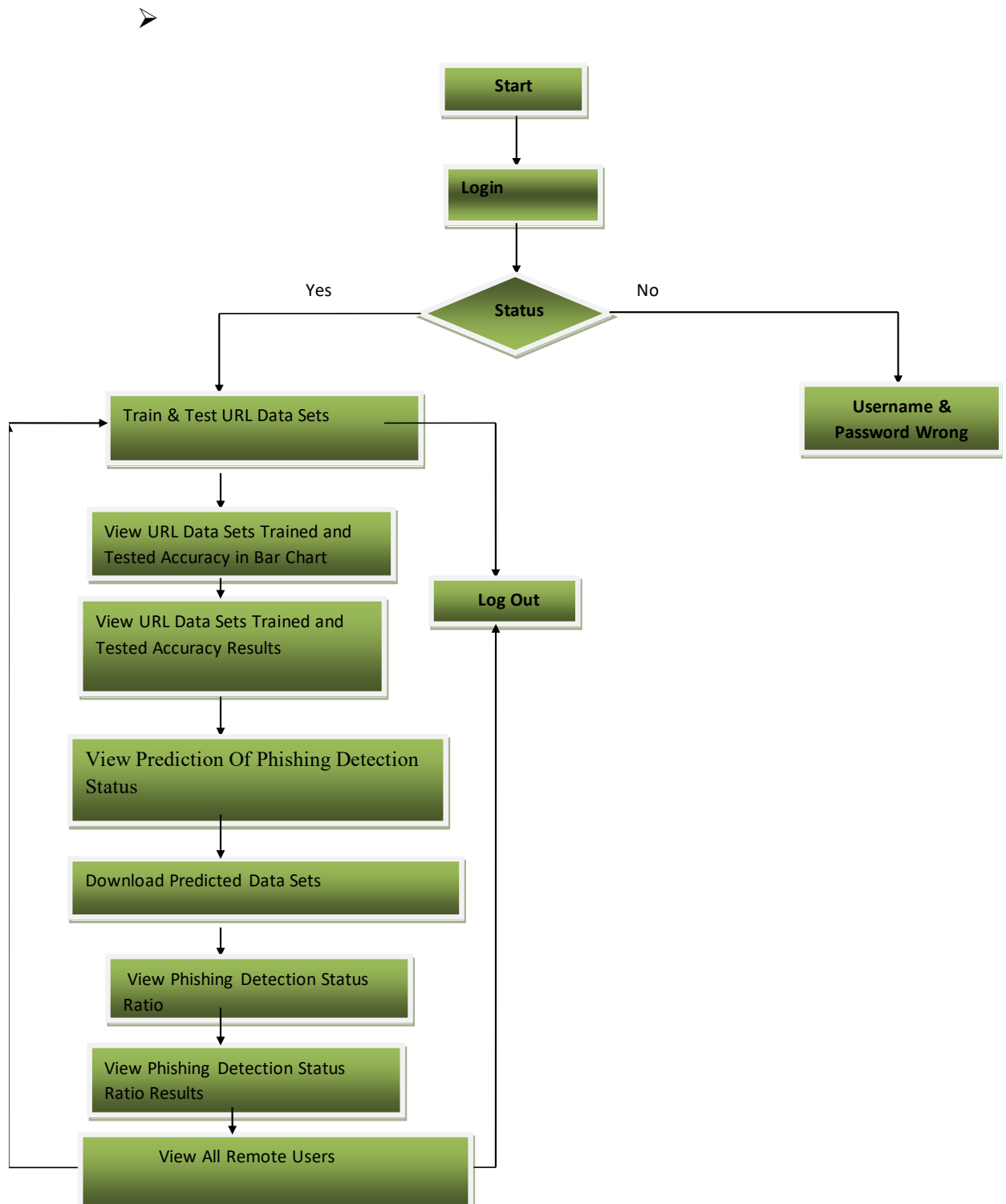


Fig:3.4.2 Flowchart For Service Provider

3.5.1 USE CASE

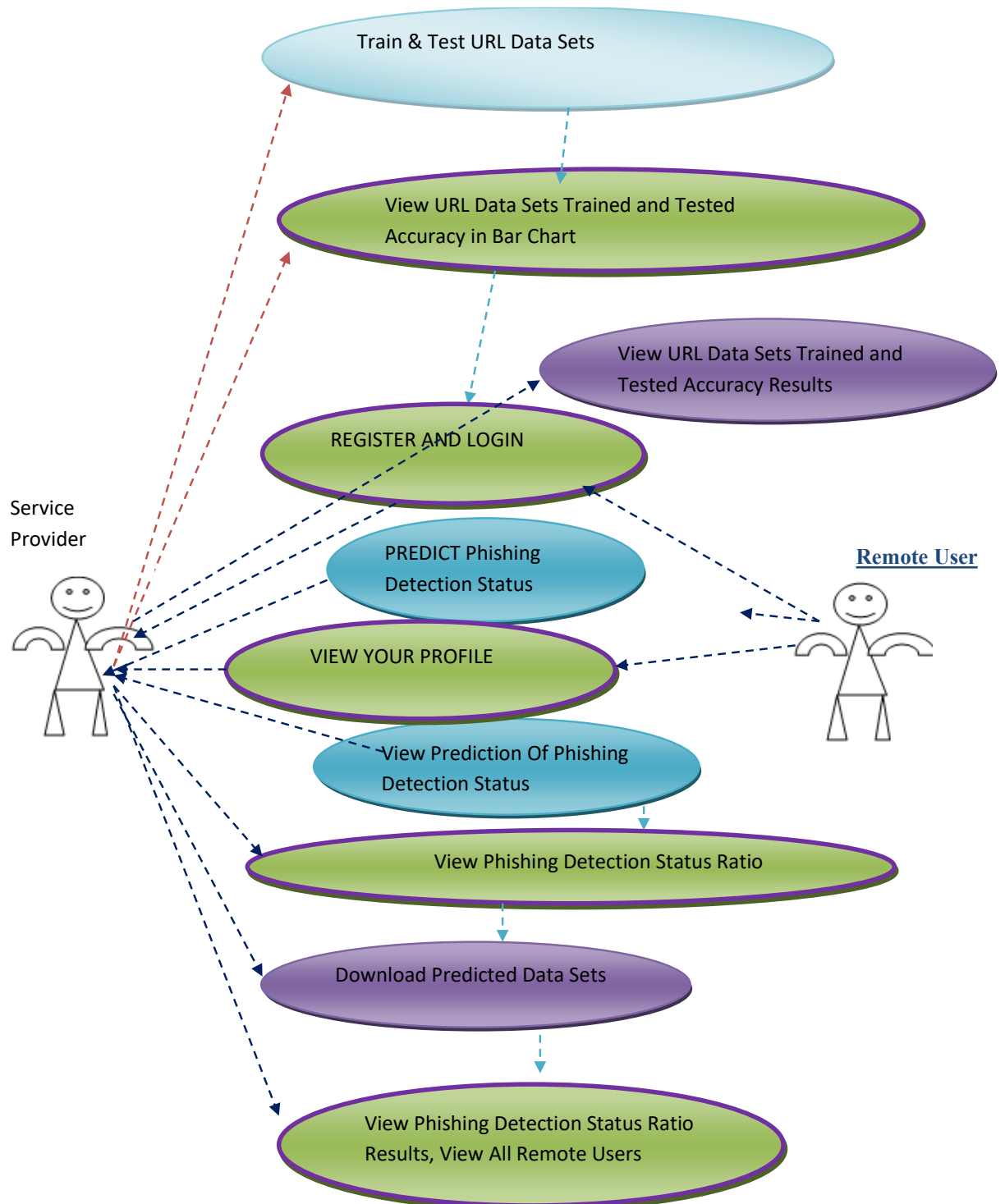


Fig:3.5.1 Use Case Diagram

3.5.2 CLASS DIAGRAM :

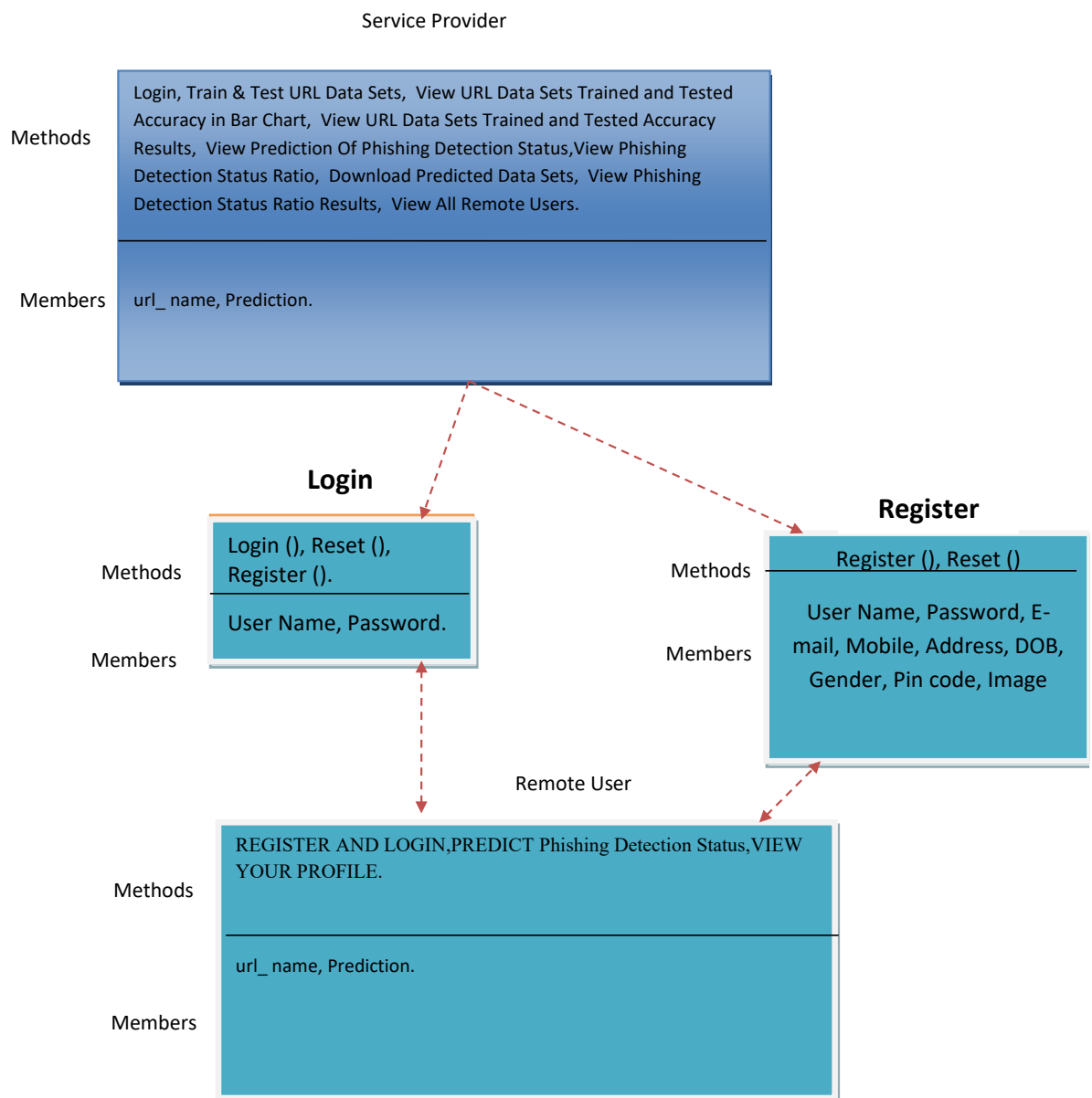


Fig:3.5.2 Class Diagram

3.5.3 SEQUENCE DIAGRAM

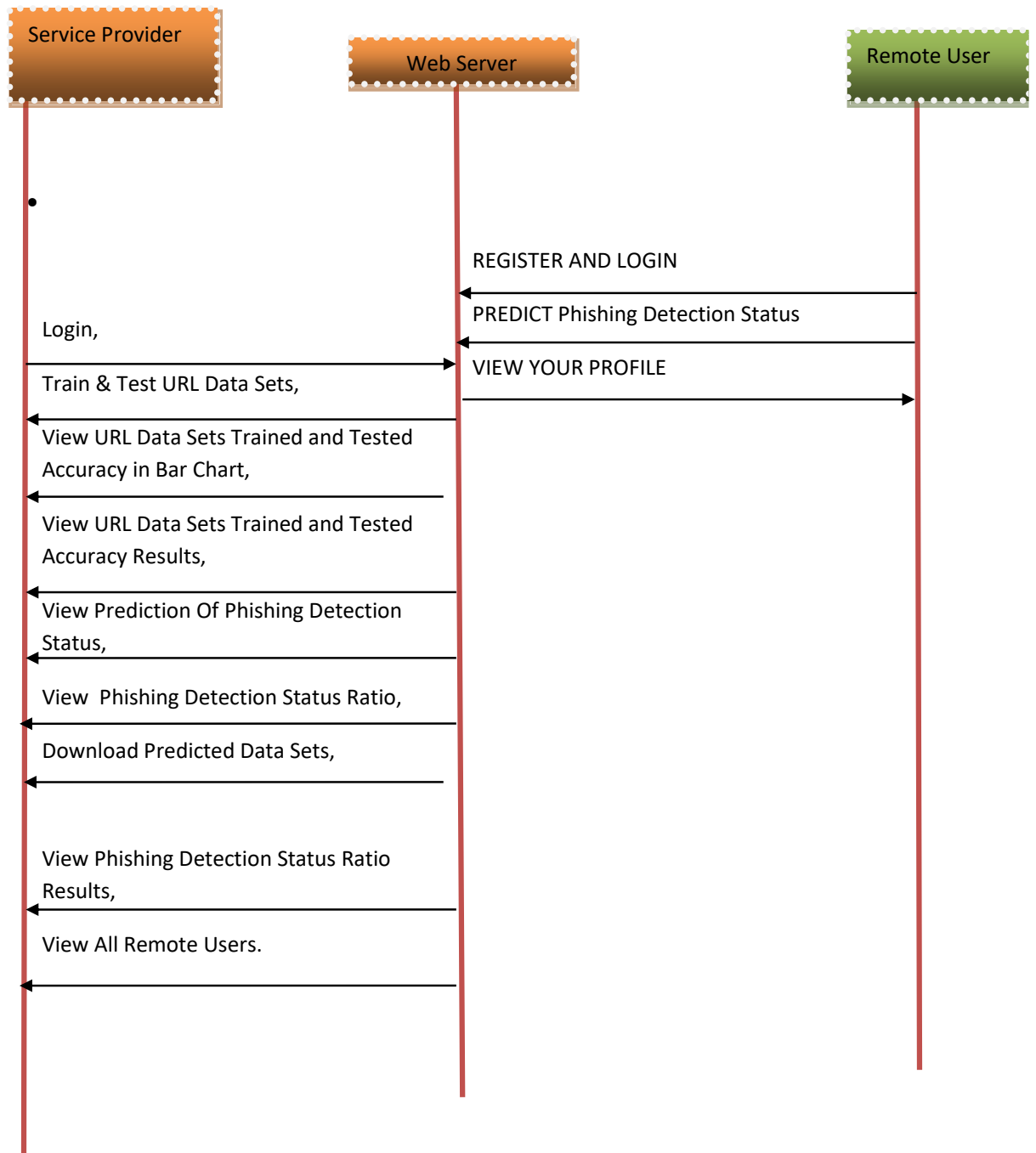


Fig:3.5.3:Sequence Diagram

CHAPTER 4

IMPLEMENTATION

4.1: MODULES:

Service provider

In this module, the Service Provider has to login by using valid user name and password. After login successful he can do some operations such as Login, Train & Test URL Data Sets, View URL Data Sets Trained and Tested Accuracy in Bar Chart, View URL Data Sets Trained and Tested Accuracy Results, View Prediction Of Phishing Detection Status, View Phishing Detection Status Ratio, Download Predicted Data Sets, View Phishing Detection Status Ratio Results, View All Remote Users.

View and Authorize Users

In this module, the admin can view the list of users who all registered. In this, the admin can view the user's details such as, user name, email, address and admin authorizes the users.

Remote User

In this module, there are n numbers of users are present. User should register before doing any operations. Once user registers, their details will be stored to the database. After registration successful, he has to login by using authorized user name and password. Once Login is successful user will do some operations like REGISTER AND LOGIN, PREDICT PHISHING DETECTION STATUS, VIEW YOUR PROFILE.

4.2: ALGORITHMS:

Decision tree classifiers

Decision tree classifiers are used successfully in many diverse areas. Their most important feature is the capability of capturing descriptive decision making knowledge from the supplied data. Decision tree can be generated from training sets. The procedure for such generation based on the set of objects (S), each belonging to one of the classes $C_1, C_2, \dots, C_k$ is as follows:

Step 1. If all the objects in S belong to the same class, for example C_i , the decision tree for S consists of a leaf labeled with this class

Step 2. Otherwise, let T be some test with possible outcomes $O_1, O_2, \dots, O_n$. Each object in S has one outcome for T so the test partitions S into subsets $S_1, S_2, \dots, S_n$ where each object in S_i has outcome O_i for T. T becomes the root of the decision tree and for each outcome O_i we build a subsidiary decision tree by invoking the same procedure recursively on the set S_i .

Gradient boosting:

Gradient boosting is a machine learning technique used in regression and classification tasks, among others. It gives a prediction model in the form of an ensemble of weak prediction models, which are typically decision trees. When a decision tree is the weak learner, the resulting algorithm is called gradient-boosted trees; it usually outperforms random forest. A gradient-boosted trees model is built in a stage-wise fashion as in other boosting methods, but it generalizes the other methods by allowing optimization of an arbitrary differentiable loss function.

K-Nearest Neighbors (KNN):

- Simple, but a very powerful classification algorithm
- Classifies based on a similarity measure
- Non-parametric

- Lazy learning
- Does not “learn” until the test example is given
- Whenever we have a new data to classify, we find its K-nearest neighbors from the training data

Example

- Training dataset consists of k-closest examples in feature space
- Feature space means, space with categorization variables (non-metric variables)
- Learning based on instances, and thus also works lazily because instance close to the input vector for test or prediction may take time to occur in the training dataset

Logistic regression Classifiers:

Logistic regression analysis studies the association between a categorical dependent variable and a set of independent (explanatory) variables. The name *logistic regression* is used when the dependent variable has only two values, such as 0 and 1 or Yes and No. The name *multinomial logistic regression* is usually reserved for the case when the dependent variable has three or more unique values, such as Married, Single, Divorced, or Widowed. Although the type of data used for the dependent variable is different from that of multiple regression, the practical use of the procedure is similar.

Logistic regression competes with discriminant analysis as a method for analyzing categorical-response variables. Many statisticians feel that logistic regression is more versatile and better suited for modeling most situations than is discriminant analysis. This is because logistic regression does not assume that the independent variables are normally distributed, as discriminant analysis does.

This program computes binary logistic regression and multinomial logistic regression on both numeric and categorical independent

variables. It reports on the regression equation as well as the goodness of fit, odds ratios, confidence limits, likelihood, and deviance. It performs a comprehensive residual analysis including diagnostic residual reports and plots. It can perform an independent variable subset selection search, looking for the best regression model with the fewest independent variables. It provides confidence intervals on predicted values and provides ROC curves to help determine the best cutoff point for classification. It allows you to validate your results by automatically classifying rows that are not used during the analysis.

Naïve Bayes:

The naive bayes approach is a supervised learning method which is based on a simplistic hypothesis: it assumes that the presence (or absence) of a particular feature of a class is unrelated to the presence (or absence) of any other feature.

Yet, despite this, it appears robust and efficient. Its performance is comparable to other supervised learning techniques. Various reasons have been advanced in the literature. In this tutorial, we highlight an explanation based on the representation bias. The naive bayes classifier is a linear classifier, as well as linear discriminant analysis, logistic regression or linear SVM (support vector machine). The difference lies on the method of estimating the parameters of the classifier (the learning bias).

While the Naive Bayes classifier is widely used in the research world, it is not widespread among practitioners which want to obtain usable results. On the one hand, the researchers found especially it is very easy to program and implement it, its parameters are easy to estimate, learning is very fast even on very large databases, its accuracy is reasonably good in comparison to the other approaches. On the other hand, the final users do not obtain a model easy to interpret and deploy, they does not understand the interest of such a technique.

Thus, we introduce in a new presentation of the results of the learning process. The classifier is easier to understand, and its deployment is also made easier. In the first part of this tutorial, we present some theoretical aspects of the naive bayes classifier. Then, we implement the approach on a dataset with Tanagra. We compare the obtained results (the parameters of the model) to those obtained with other linear approaches such as the logistic regression, the linear discriminant analysis and the linear SVM. We note that the results are highly consistent. This largely explains the good performance of the method in comparison to others. In the second part, we use various tools on the same dataset (Weka 3.6.0, R 2.9.2, Knime 2.1.1, Orange 2.0b and RapidMiner 4.6.0). We try above all to understand the obtained results.

Random Forest:

Random forests or random decision forests are an ensemble learning method for classification, regression and other tasks that operates by constructing a multitude of decision trees at training time. For classification tasks, the output of the random forest is the class selected by most trees. For regression tasks, the mean or average prediction of the individual trees is returned. Random decision forests correct for decision trees' habit of overfitting to their training set. Random forests generally outperform decision trees, but their accuracy is lower than gradient boosted trees. However, data characteristics can affect their performance.

The first algorithm for random decision forests was created in 1995 by Tin Kam Ho using the random subspace method, which, in Ho's formulation, is a way to implement the "stochastic discrimination" approach to classification proposed by Eugene Kleinberg.

An extension of the algorithm was developed by Leo Breiman and Adele Cutler, who registered "Random Forests" as a trademark in 2006 (as of 2019, owned by Minitab, Inc.). The extension combines Breiman's "bagging" idea and random selection of features, introduced first by

Ho[1] and later independently by Amit and Geman[13] in order to construct a collection of decision trees with controlled variance.

Random forests are frequently used as "blackbox" models in businesses, as they generate reasonable predictions across a wide range of data while requiring little configuration.

SVM:

In classification tasks a discriminant machine learning technique aims at finding, based on an *independent and identically distributed (iid)* training dataset, a discriminant function that can correctly predict labels for newly acquired instances. Unlike generative machine learning approaches, which require computations of conditional probability distributions, a discriminant classification function takes a data point x and assigns it to one of the different classes that are a part of the classification task. Less powerful than generative approaches, which are mostly used when prediction involves outlier detection, discriminant approaches require fewer computational resources and less training data, especially for a multidimensional feature space and when only posterior probabilities are needed. From a geometric perspective, learning a classifier is equivalent to finding the equation for a multidimensional surface that best separates the different classes in the feature space.

SVM is a discriminant technique, and, because it solves the convex optimization problem analytically, it always returns the same optimal hyperplane parameter—in contrast to *genetic algorithms (GAs)* or *perceptrons*, both of which are widely used for classification in machine learning. For perceptrons, solutions are highly dependent on the initialization and termination criteria. For a specific kernel that transforms the data from the input space to the feature space, training returns uniquely defined SVM model parameters for a given training

set, whereas the perceptron and GA classifier models are different each time training is initialized. The aim of GAs and perceptrons is only to minimize error during training, which will translate into several hyperplanes' meeting this requirement.

CHAPTER-5

RESULTS

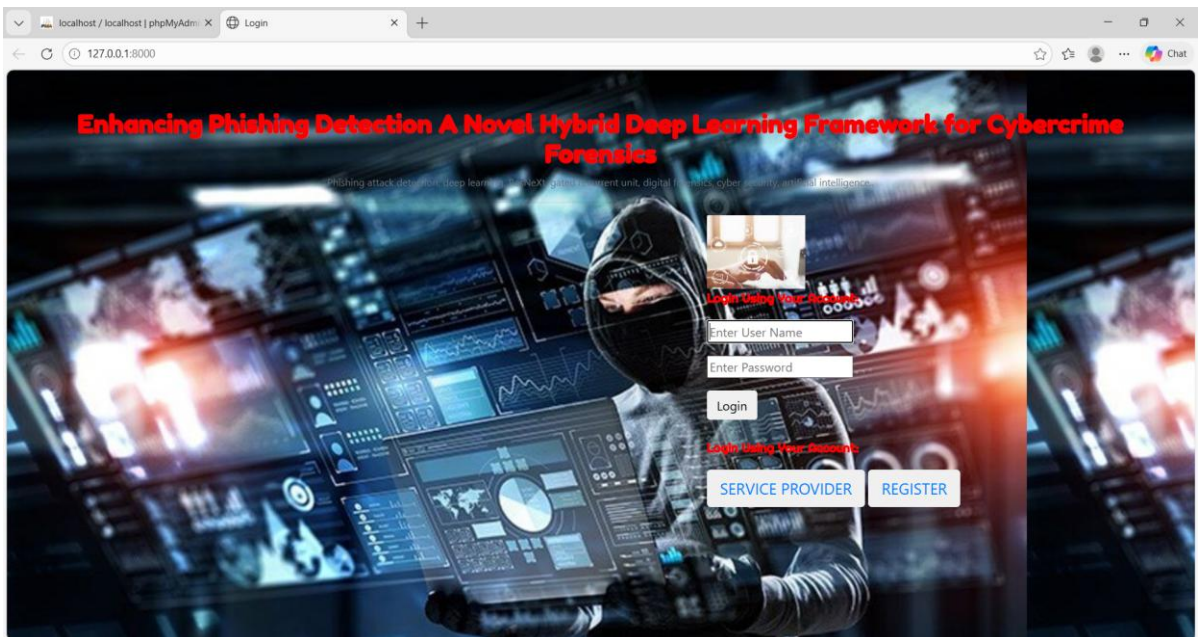


Fig:5.1 Home page

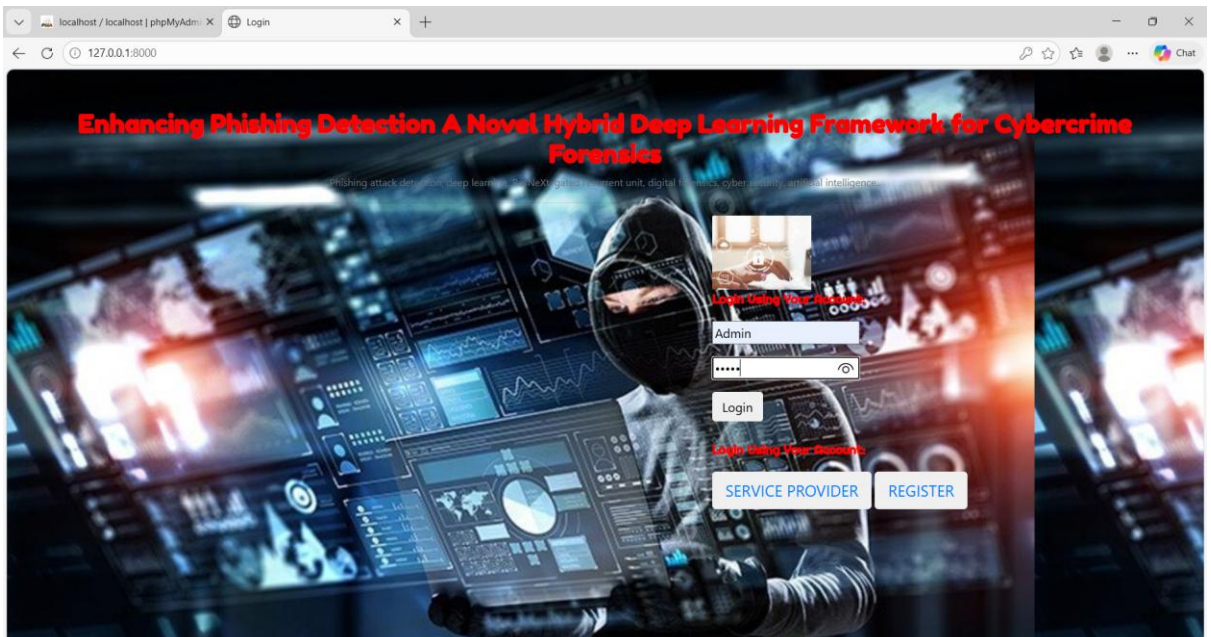


Fig:5.2 Service Provider login

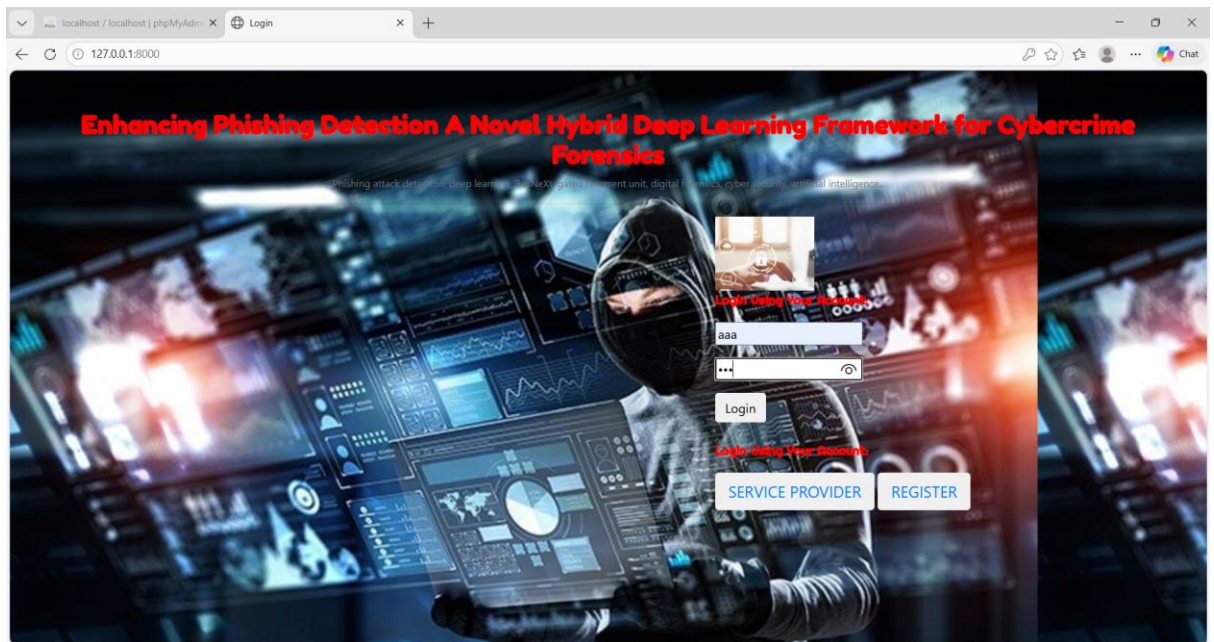


Fig:5.3 User login

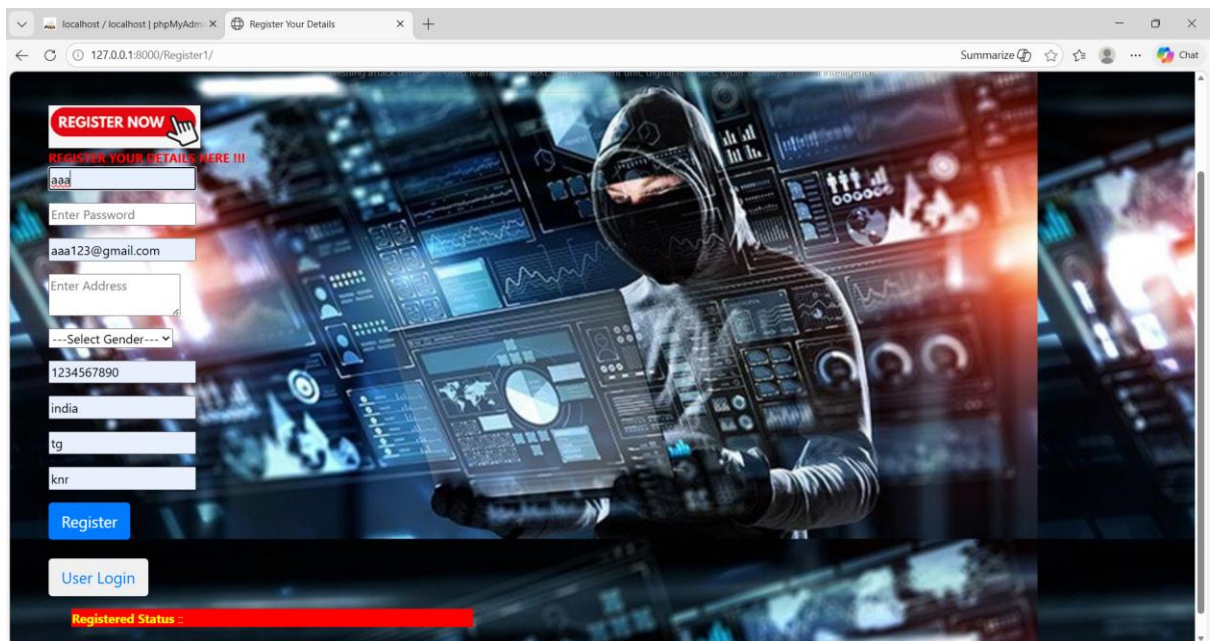


Fig:5.4 New User Login Registration

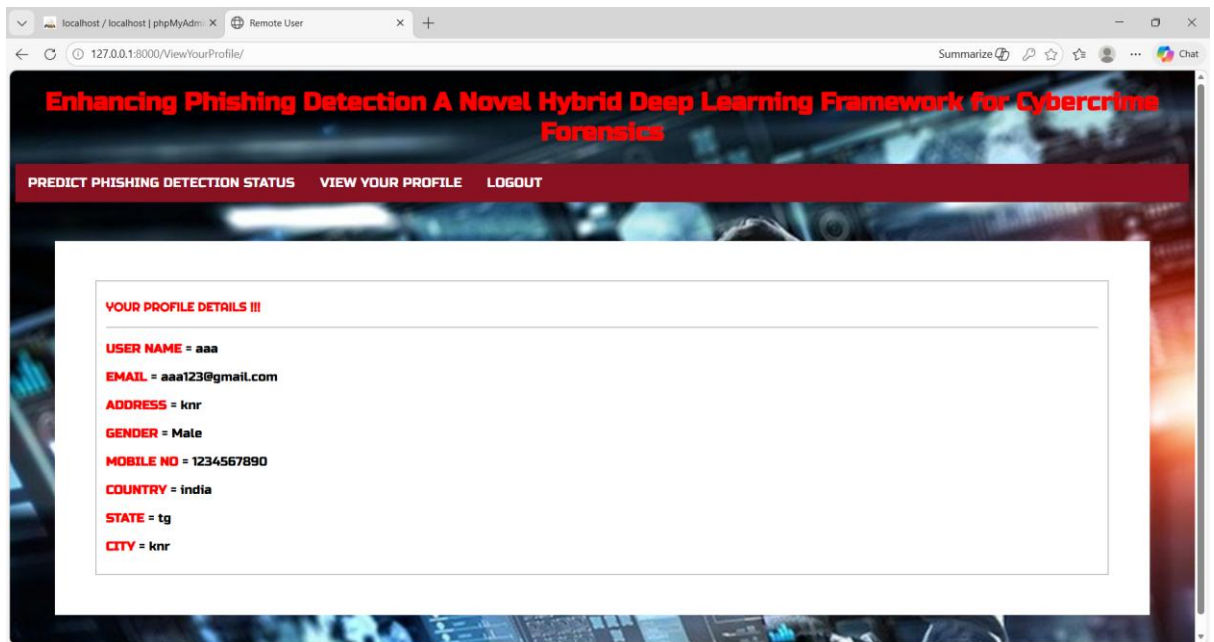


Fig:5.5 Profile Details

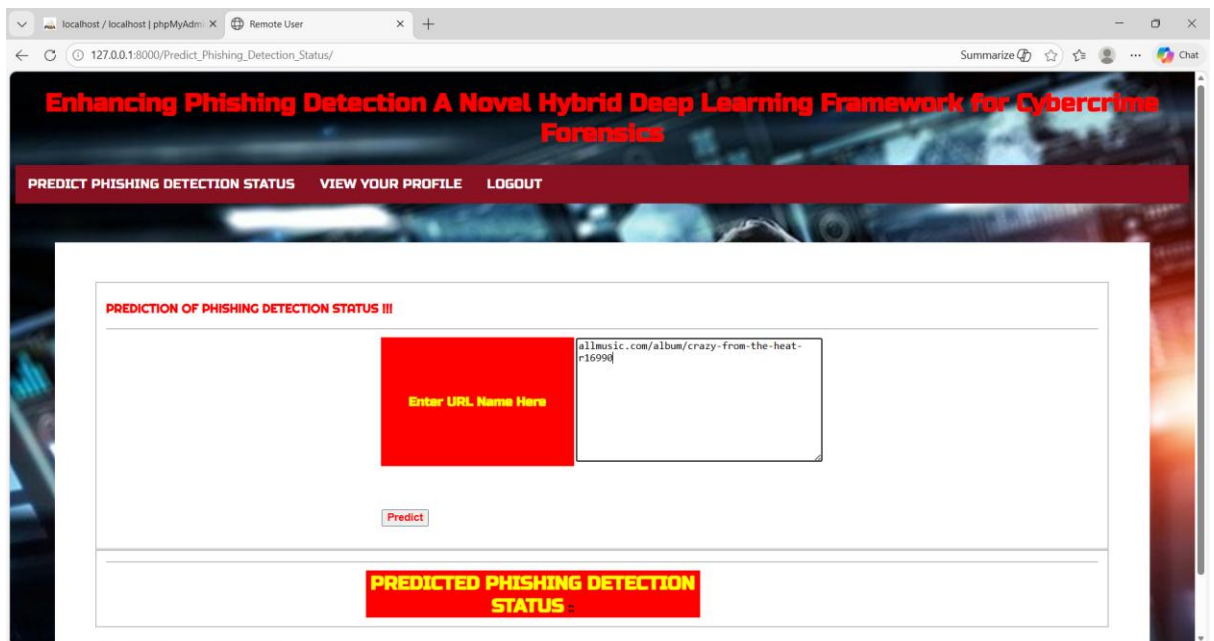


Fig:5.6 Predication Type

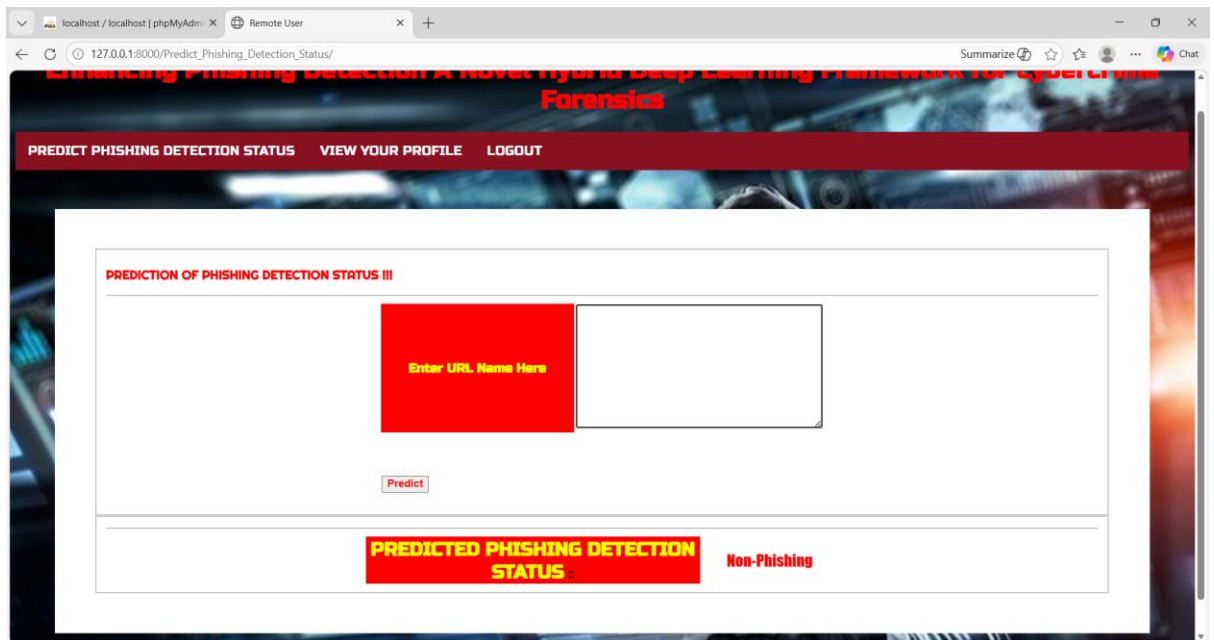


Fig:5.7 Successfully Predicted

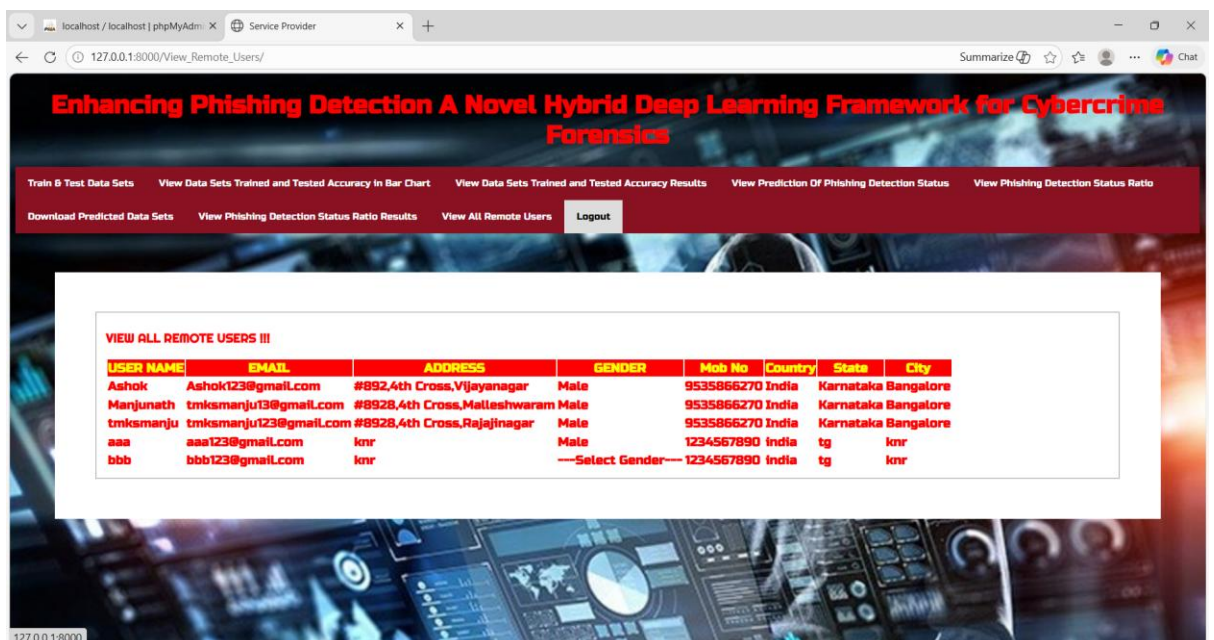


Fig:5.8 View All Remotes Users

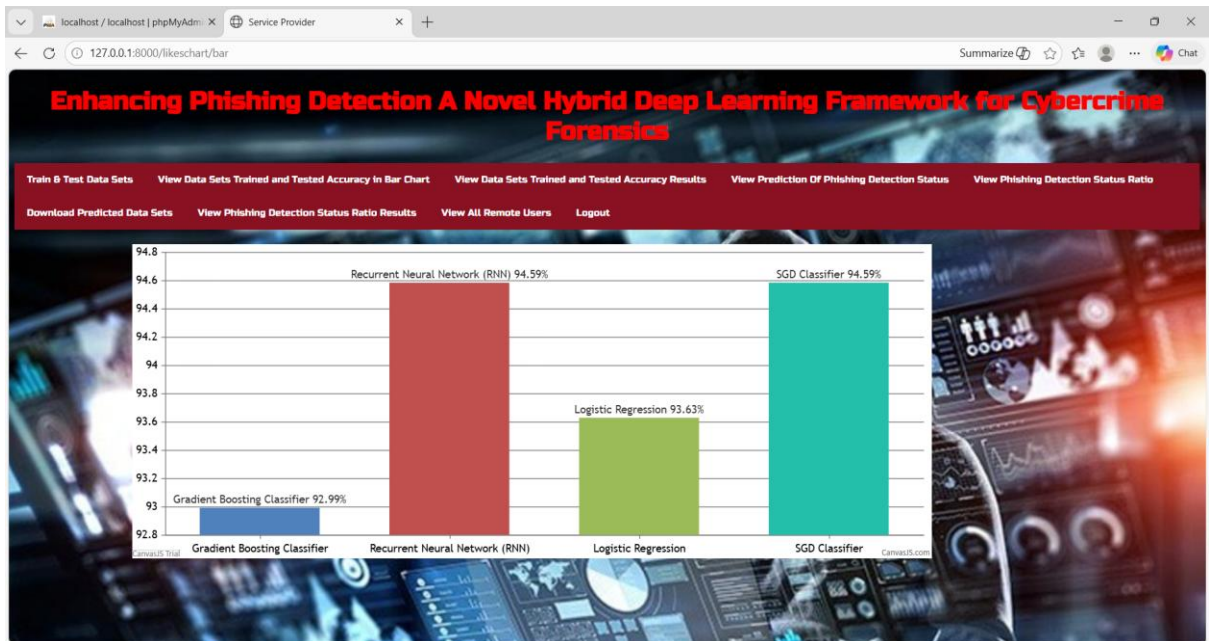


Fig:5.9 Bar Chart

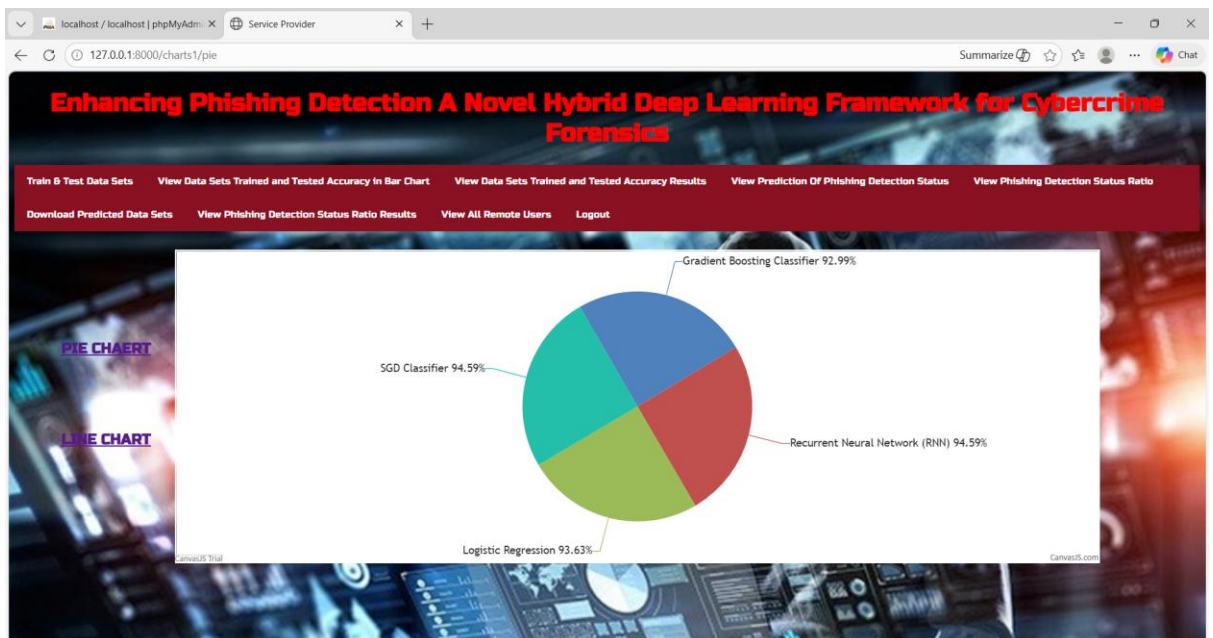


Fig:5.10 Pie Chart

Train & Test Data Sets View Data Sets Trained and Tested Accuracy in Bar Chart View Data Sets Trained and Tested Accuracy Results View Prediction Of Phishing Detection Status View Phishing Detection Status Ratio

Download Predicted Data Sets View Phishing Detection Status Ratio Results View All Remote Users Logout

View Phishing Detection Status Details III

Phishing Detection Status	Prediction Type
allmusic.com/album/crazy-from-the-heat-r16990	Non-Phishing
http://portal.dddgaming.com/docs/rules/15027/cn/game_cn.html?amiuMjAxNQ%3D%3D	Malware
http://mmc-leipzig.com/index.php	Defacement
192.com/atoz/people/sturgeon/craig/	Non-Phishing
tophyipsites.com	Non-Phishing
http://update-information001.alhayan.ly/	Phishing
http://42.227.166.214:52835/Mozi.m	Malware
http://lorsiris.com/index.php?option=com_jevents&task=day.listevents&year=2013&month=01&day=25&Itemid=50	Defacement
aol.sportingnews.com/ncaa-basketball/feed/2011-09/top-50/story/no-9-baylor-bears-glance	Non-Phishing

Fig:5.11 Phishing Detection Predication Type Details

CHAPTER-6

TESTING

SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

TYPES OF TESTS

UNIT TESTING

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

INTEGRATION TESTING

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successful unit testing, the combination of components is correct and consistent. Integration

testing is specifically aimed at exposing the problems that arise from the combination of components.

FUNCTIONAL TEST

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures: interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

SYSTEM TEST

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

WHITE BOX TESTING

White Box Testing is a testing in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is used to test areas that cannot be reached from a black box level.

BLACK BOX TESTING

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

6.1 UNIT TESTING:

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

TEST STRATEGY AND APPROACH

Field testing will be performed manually and functional tests will be written in detail.

TEST OBJECTIVES

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

FEATURES TO BE TESTED

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

6.2 INTEGRATION TESTING

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

6.3 ACCEPTANCE TESTING

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

SYSTEM TESTING

TESTING METHODOLOGIES

The following are the Testing Methodologies:

- **Unit Testing.**
- **Integration Testing.**
- **User Acceptance Testing.**
- **Output Testing.**
- **Validation Testing.**

UNIT TESTING

Unit testing focuses verification effort on the smallest unit of Software design that is the module. Unit testing exercises specific paths in a module's control structure to ensure complete coverage and maximum error detection. This test focuses on each module individually, ensuring that it functions properly as a unit. Hence, the naming is Unit Testing.

During this testing, each module is tested individually and the module interfaces are verified for the consistency with design

specification. All important processing path are tested for the expected results. All error handling paths are also tested.

INTEGRATION TESTING

Integration testing addresses the issues associated with the dual problems of verification and program construction. After the software has been integrated a set of high order tests are conducted. The main objective in this testing process is to take unit tested modules and builds a program structure that has been dictated by design.

The following are the types of Integration Testing:

1)Top Down Integration

This method is an incremental approach to the construction of program structure. Modules are integrated by moving downward through the control hierarchy, beginning with the main program module. The module subordinates to the main program module are incorporated into the structure in either a depth first or breadth first manner.

In this method, the software is tested from main module and individual stubs are replaced when the test proceeds downwards.

2. Bottom-up Integration

This method begins the construction and testing with the modules at the lowest level in the program structure. Since the modules are integrated from the bottom up, processing required for modules subordinate to a given level is always available and the need for stubs is eliminated. The bottom up integration strategy may be implemented with the following steps:

- The low-level modules are combined into clusters into clusters that perform a specific Software sub-function.
- A driver (i.e.) the control program for testing is written to coordinate test case input and output.
- The cluster is tested.

- Drivers are removed and clusters are combined moving upward in the program structure

The bottom up approaches tests each module individually and then each module is integrated with a main module and tested for functionality.

OTHER TESTING METHODOLOGIES

USER ACCEPTANCE TESTING

User Acceptance of a system is the key factor for the success of any system. The system under consideration is tested for user acceptance by constantly keeping in touch with the prospective system users at the time of developing and making changes wherever required. The system developed provides a friendly user interface that can easily be understood even by a person who is new to the system.

OUTPUT TESTING

After performing the validation testing, the next step is output testing of the proposed system, since no system could be useful if it does not produce the required output in the specified format. Asking the users about the format required by them tests the outputs generated or displayed by the system under consideration. Hence the output format is considered in 2 ways – one is on screen and another in printed format.

VALIDATION CHECKING

Validation checks are performed on the following fields.

TEXT FIELD:

The text field can contain only the number of characters lesser than or equal to its size. The text fields are alphanumeric in some tables and alphabetic in other tables. Incorrect entry always flashes and error message.

NUMERIC FIELD:

The numeric field can contain only numbers from 0 to 9. An entry of any character flashes an error messages. The individual modules are checked for accuracy and what it has to perform. Each module is subjected to test run along with sample data. The individually tested modules are integrated into a single system. Testing involves executing the real data information is used in the program the existence of any program defect is inferred from the output. The testing should be planned so that all the requirements are individually tested.

A successful test is one that gives out the defects for the inappropriate data and produces and output revealing the errors in the system.

PREPARATION OF TEST DATA

Taking various kinds of test data does the above testing. Preparation of test data plays a vital role in the system testing. After preparing the test data the system under study is tested using that test data. While testing the system by using test data errors are again uncovered and corrected by using above testing steps and corrections are also noted for future use.

USING LIVE TEST DATA:

Live test data are those that are actually extracted from organization files. After a system is partially constructed, programmers or analysts often ask users to key in a set of data from their normal activities. Then, the systems person uses this data as a way to partially test the system. In other instances, programmers or analysts extract a set of live data from the files and have them entered themselves.

It is difficult to obtain live data in sufficient amounts to conduct extensive testing. And, although it is realistic data that will show how the system will perform for the typical processing requirement, assuming that the live data entered are in fact typical, such data generally will not test all combinations or formats that can enter the

system. This bias toward typical values then does not provide a true systems test and in fact ignores the cases most likely to cause system failure.

USING ARTIFICIAL TEST DATA:

Artificial test data are created solely for test purposes, since they can be generated to test all combinations of formats and values. In other words, the artificial data, which can quickly be prepared by a data generating utility program in the information systems department, make possible the testing of all login and control paths through the program.

The most effective test programs use artificial test data generated by persons other than those who wrote the programs. Often, an independent team of testers formulates a testing plan, using the systems specifications.

The package “Virtual Private Network” has satisfied all the requirements specified as per software requirement specification and was accepted.

USER TRAINING

Whenever a new system is developed, user training is required to educate them about the working of the system so that it can be put to efficient use by those for whom the system has been primarily designed. For this purpose the normal working of the project was demonstrated to the prospective users. Its working is easily understandable and since the expected users are people who have good knowledge of computers, the use of this system is very easy.

MAINTAINENCE

This covers a wide range of activities including correcting code and design errors. To reduce the need for maintenance in the long run, we have more accurately defined the user’s requirements during the process of system development. Depending on the requirements, this system has been developed to satisfy the needs to the largest possible extent. With development in technology, it may be possible to add many more features based on the requirements in future. The coding and

designing is simple and easy to understand which will make maintenance easier.

TESTING STRATEGY :

A strategy for system testing integrates system test cases and design techniques into a well planned series of steps that results in the successful construction of software. The testing strategy must co-operate test planning, test case design, test execution, and the resultant data collection and evaluation .A strategy for software testing must accommodate low-level tests that are necessary to verify that a small source code segment has been correctly implemented as well as high level tests that validate major system functions against user requirements.

Software testing is a critical element of software quality assurance and represents the ultimate review of specification design and coding. Testing represents an interesting anomaly for the software. Thus, a series of testing are performed for the proposed system before the system is ready for user acceptance testing.

SYSTEM TESTING:

Software once validated must be combined with other system elements (e.g. Hardware, people, database). System testing verifies that all the elements are proper and that overall system function performance is achieved. It also tests to find discrepancies between the system and its original objective, current specifications and system documentation.

UNIT TESTING:

In unit testing different are modules are tested against the specifications produced during the design for the modules. Unit testing is essential for verification of the code produced during the coding phase, and hence the goals to test the internal logic of the modules. Using the detailed design description as a guide, important Conrail paths are tested to uncover errors within the boundary of the modules. This testing is carried out during the programming stage itself. In this

type of testing step, each module was found to be working satisfactorily as regards to the expected output from the module.

In Due Course, latest technology advancements will be taken into consideration. As part of technical build-up many components of the networking system will be generic in nature so that future projects can either use or interact with this. The future holds a lot to offer to the development and refinement of this project.

CHAPTER-7

CONCLUSION

With significant implications for digital forensics and online security, our work represents a breakthrough in the ongoing problem of phishing website detection. Despite technical advancements, creative solutions are still necessary to counter growing phishing threats. Our model RNT-J, specifically designed for real-time phishing website analysis within the digital forensics framework, handles modern complexity and big datasets related to phishing threats with exceptional performance. RNT-J is better able to identify and assess phishing behaviour in addition to digital forensics. In a comparative analysis against ResNet, DenseNet, BERT, and ELMo, RNT-J outperforms these methods in terms of accuracy, swift pattern identification, and efficient integration with digital forensics capabilities. Two key elements of RNTJ's superiority are the considerable improvement in phishing website recognition accuracy and the quick discovery of hitherto unknown patterns. This concept provides a comprehensive strategy that effectively addresses inefficiencies in the present methods employed in the field of digital forensics. Our model aptly reflects the need for more comprehensive comparative studies, demonstrating the potential of RNT-J for performance evaluations based on real-time information obtained from phishing websites. This methodology, when seen within the broader context of digital forensics, represents a major advancement in the fight against phishing attempts, leading to improved security and operational efficacy.

Our project contributes significantly to defence online activities in the field of cyber security, where sophisticated algorithms are necessary to fortify defences. It also closely studies digital crimes associated with phishing attempts. We want to use more sophisticated hybrid algorithms in the future, which will improve the accuracy and efficiency with which we identify phishing attacks.

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